

量子力学期末考试参考答案

第一题 C C B A D

第二题 证明题，略

第三题 $E_n = \frac{Z^2}{n^2} Ry, \quad Ry = \frac{1}{2} \alpha^2 m_e c^2$

1.
$$\Psi(\vec{r}, t) = \frac{1}{\sqrt{11}} \left[\sqrt{3} \psi_{2,1,-1}(\vec{r}) - \psi_{2,1,0}(\vec{r}) + \sqrt{5} \psi_{2,1,1}(\vec{r}) \right] e^{-\frac{i}{\hbar} E_2 t} + \sqrt{\frac{2}{11}} \psi_{3,1,1}(\vec{r}) e^{-\frac{i}{\hbar} E_3 t}$$

2. 测量 E_2 的几率为 $\frac{9}{11}$, 测量 E_3 的几率为 $\frac{2}{11}$

3. 测量 L_z 获得 $-1\hbar$ 的几率为 $\frac{3}{11}$

第四题

$$|\langle \mathbf{m} | \mathbf{n}; \pm \rangle|^2 = \frac{1}{2}(1 \pm \mathbf{n} \cdot \mathbf{m})$$

$$|\mathbf{n}; +\rangle = \cos \frac{\theta}{2} |+\rangle_z + \sin \frac{\theta}{2} e^{i\phi} |-\rangle_z$$

$$|\mathbf{n}; -\rangle = -\sin \frac{\theta}{2} e^{-i\phi} |+\rangle_z + \cos \frac{\theta}{2} |-\rangle_z$$

考虑 $\mathbf{n}$ 的正极化方向 $|\mathbf{n}; +\rangle$, 根据题意,

$$\frac{1}{2}(1 + \cos \theta) = \frac{1}{3} \quad \frac{1}{2}(1 + \sin \theta \cos \phi) = \frac{1}{6}$$

$$\cos \frac{\theta}{2} = \frac{1}{\sqrt{3}}, \quad \sin \frac{\theta}{2} = \sqrt{\frac{2}{3}}, \quad \cos \phi = -\frac{1}{\sqrt{2}} \rightarrow \phi = \frac{3\pi}{4}$$

S_y 的平均值为: $\langle \mathbf{n}; + | S_y | \mathbf{n}, + \rangle = \frac{\hbar}{2} \langle \mathbf{n}; + | \sigma_y | \mathbf{n}, + \rangle = \frac{\hbar}{2} \sin \theta \sin \phi = \frac{\hbar}{3}$

同样, $\mathbf{n}$ 的负极化方向 $|\mathbf{n}; -\rangle$, 可以得到,

$$\sin \frac{\theta}{2} = \frac{1}{\sqrt{3}}, \quad \cos \frac{\theta}{2} = \sqrt{\frac{2}{3}}, \quad \cos \phi = \frac{1}{\sqrt{2}} \rightarrow \phi = \frac{\pi}{4}$$

$$\langle \mathbf{n}; - | S_y | \mathbf{n}, - \rangle = \frac{\hbar}{2} \langle \mathbf{n}; - | \sigma_y | \mathbf{n}, - \rangle = -\frac{\hbar}{2} \sin \theta \sin \phi = -\frac{\hbar}{3}$$

第五题

$$\begin{aligned} & A(S_1 \cdot S_2 - 3S_{1z}S_{2z})|SM\rangle \\ &= \frac{A}{2}(S^2 - S_1^2 - S_2^2)|SM\rangle - \frac{3A}{2}(S_z^2 - S_{1z}^2 - S_{2z}^2)|SM\rangle \\ &= \frac{A}{2}\left[S(S+1) - \frac{1}{2} \times \frac{3}{2} - \frac{1}{2} \times \frac{3}{2}\right]\hbar^2|SM\rangle - \frac{3A}{2}\left(M^2 - \frac{1}{4} - \frac{1}{4}\right)\hbar^2|SM\rangle \\ &= \frac{A}{2}[S(S+1) - 3M^2]\hbar^2|SM\rangle \end{aligned}$$

$ SM\rangle$	本征值
$ 00\rangle$	0
$ 11\rangle$	$-\frac{1}{2}A\hbar^2$
$ 10\rangle$	$1A\hbar^2$
$ 1,-1\rangle$	$-\frac{1}{2}A\hbar^2$

第六题

$$\begin{aligned}
 -\frac{e}{2m_e c} \mathbf{B} \cdot \left\langle \ell m; \frac{1}{2} m_s \left| \mathbf{L} \right| \ell m; \frac{1}{2} m'_s \right\rangle &= -\frac{e}{2m_e c} \delta_{m_s m'_s} \mathbf{B} \cdot \left\langle \ell m; \frac{1}{2} m_s \left| \mathbf{L} \right| \ell m; \frac{1}{2} m'_s \right\rangle \\
 &= -\frac{m \hbar e}{2m_e c} \delta_{m_s m'_s} B_z
 \end{aligned}$$

$$\begin{aligned}
 &-\frac{e}{m_e c} \mathbf{B} \cdot \left\langle \ell m; \frac{1}{2} m_s \left| \mathbf{S} \right| \ell m; \frac{1}{2} m'_s \right\rangle \\
 &= -\frac{e \hbar}{2m_e c} \begin{pmatrix} B_z & B_x - iB_y \\ B_x + iB_y & -B_z \end{pmatrix}_{m_s m'_s} \quad \longrightarrow \quad \begin{aligned}
 &\left\langle \ell m; \frac{1}{2} \uparrow \left| H_M \right| \ell m; \frac{1}{2} \uparrow \right\rangle = -\frac{(m+1)\hbar e}{2m_e} B_z \\
 &\left\langle \ell m; \frac{1}{2} \downarrow \left| H_M \right| \ell m; \frac{1}{2} \downarrow \right\rangle = -\frac{(m-1)\hbar e}{2m_e} B_z \\
 &\left\langle \ell m; \frac{1}{2} \uparrow \left| H_M \right| \ell m; \frac{1}{2} \downarrow \right\rangle = \left\langle \ell m; \frac{1}{2} \downarrow \left| H_M \right| \ell m; \frac{1}{2} \uparrow \right\rangle^* \\
 &= -\frac{\hbar e}{2m_e} (B_x - iB_y)
 \end{aligned}
 \end{aligned}$$

$$\begin{aligned}
 \left\langle \ell m; \frac{1}{2} m_s \left| \mathbf{r} \right| \ell m; \frac{1}{2} m'_s \right\rangle &= -\left\langle \ell m; \frac{1}{2} m_s \left| \mathcal{P} \mathbf{r} \mathcal{P} \right| \ell m; \frac{1}{2} m'_s \right\rangle \\
 &= -(-1)^{2\ell} \left\langle \ell m; \frac{1}{2} m_s \left| \mathbf{r} \right| \ell m; \frac{1}{2} m'_s \right\rangle \quad \longrightarrow \quad \left\langle \ell, m; \frac{1}{2}, m_s \left| H_E \right| \ell, m; \frac{1}{2}, m'_s \right\rangle = 0 \\
 &= -\left\langle \ell m; \frac{1}{2} m_s \left| \mathbf{r} \right| \ell m; \frac{1}{2} m'_s \right\rangle = 0
 \end{aligned}$$

第七题

$$\psi_n(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right), \quad E_n = \frac{\hbar^2 \pi^2 n^2}{2ma^2}, \quad n = 1, 2, \dots$$

$$\begin{aligned} \Psi_{n,m}^S &= \frac{1}{\sqrt{2}} [\psi_n(x_1)\psi_m(x_2) + \psi_m(x_1)\psi_n(x_2)] & E_{n,m} &= E_n^1 + E_m^2 = \frac{\hbar^2 \pi^2 (n^2 + m^2)}{2ma^2} \\ \Psi_{n,m}^A &= \frac{1}{\sqrt{2}} [\psi_n(x_1)\psi_m(x_2) - \psi_m(x_1)\psi_n(x_2)] \end{aligned}$$

- 1. 自旋三重态 χ_1 ，空间反称 $\Psi_{1,2}^A$ ，能量为 $E_{1,2}$ ，状态为 $|T\rangle = |\Psi_{1,2}^A\rangle \otimes |\chi_1\rangle$
- 2. 自旋单态 χ_0 ，空间对称 $\Psi_{1,1}^S$ ，能量为 $E_{1,1}$ ，状态为 $|S\rangle = |\Psi_{1,1}^S\rangle \otimes |\chi_0\rangle$
- 3. 对自旋三重态 χ_1 ，由于空间反称 $\Psi_{1,2}^A$ ，微扰能量 $\Delta E = 0$

对自旋单态，空间部分对称，微扰能量为

$$\Delta E_S = \langle S | \delta H | S \rangle = \lambda \left[\int \Psi_{1,1}^{S*}(x_1, x_2) \Psi_{1,1}^S(x_1, x_2) \delta(x_1 - x_2) dx_1 dx_2 \right] \langle \chi_0 | \mathbf{S}_1 \cdot \mathbf{S}_2 | \chi_0 \rangle$$

$$\langle \chi_0 | \mathbf{S}_1 \cdot \mathbf{S}_2 | \chi_0 \rangle = \frac{1}{2} \langle \chi_0 | (\mathbf{S}^2 - \mathbf{S}_1^2 - \mathbf{S}_2^2) | \chi_0 \rangle = -\frac{3}{4} \hbar^2$$

$$\int \Psi_{1,1}^{S*}(x_1, x_2) \Psi_{1,1}^S(x_1, x_2) \delta(x_1 - x_2) dx_1 dx_2 = \int |\Psi_{1,1}^S(x, x)|^2 dx = 2 \int_0^a \psi_1^4(x) dx$$

$$= 2 \int_0^a \left(\frac{2}{a}\right)^2 \sin^4\left(\frac{n\pi x}{a}\right) dx = \frac{8}{a} \int_0^1 \sin^4(n\pi x) dx = \frac{8}{a} \cdot \frac{3}{8} = \frac{3}{a}$$

$$\Delta E_S = \lambda \frac{3}{a} \cdot \left(-\frac{3}{4} \hbar^2\right) = -\frac{9}{4} \frac{\lambda \hbar^2}{a}$$